THE CHEBYSHEV CONSTRUCTION IN CONJUGATE DEGREE ONE

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The fact that $T_1(z) = z$ is linear does not prevent the Chebyshev construction from attaining $3n-2$ zeros. The additional roots come from the nonreal branches of $\operatorname{Im} T_n = 0$. This note gives a complete count and an explicit coefficient valid in every degree. The count is the sharp bound proved by Khavinson and Świątek [KS03] and attained in Geyer's construction [Gey08].

For a complex-valued map F, let $N(F)$ be its number of distinct zeros. A zero is nondegenerate when the real differential is invertible. The Chebyshev polynomial T_n is defined by $T_n(\cos\theta) = \cos(n\theta)$.

Theorem 1. *For every $n \geq 2$, the polynomial $p_n(z) = z - iT_n(z)/n^2$ satisfies $\#\{z : p_n(z) = \bar{z}\} = 3n-2$, and every zero of $p_n(z) - \bar{z}$ is nondegenerate.*

1. The real-level curves

The identity $T_n(\cos\zeta) = \cos(n\zeta)$ holds for complex ζ, by the polynomial recurrence for T_n. Every point of the upper half-plane has a unique representation

$$z = \cos(\theta - it) = \cos\theta\cosh t + i\sin\theta\sinh t, \qquad 0 < \theta < \pi, \quad t > 0. \tag{1}$$

Indeed, of the two solutions of $\xi + \xi^{-1} = 2z$, exactly one has modulus greater than one. Its imaginary part is positive, and writing it as $e^{t+i\theta}$ gives (1). The differential is nonsingular: its determinant is $-(\sin^2\theta + \sinh^2 t) < 0$. Since $\operatorname{Im} T_n(z) = \sin(n\theta)\sinh(nt)$, the nonreal upper branches of the real-level set are

$$z_j(t) = \cos\theta_j\cosh t + i\sin\theta_j\sinh t, \qquad \theta_j = \frac{j\pi}{n}, \quad 1 \leq j < n, \quad t > 0. \tag{2}$$

The remaining branches are their complex conjugates and the real axis. On the j-th branch, $T_n(z_j(t)) = (-1)^j\cosh(nt)$.

2. A complete count when the conjugate degree is one

For every real a, one has $\operatorname{Im} T_1(z-a) = \operatorname{Im} z$. We therefore study $\varepsilon T_n(z) - 2\operatorname{Im} z$. Put $\phi_n(t) = \sinh t/\cosh(nt)$ for $t > 0$. Its logarithmic derivative is $\coth t - n\tanh(nt)$, whose derivative is $-\operatorname{csch}^2 t - n^2\operatorname{sech}^2(nt) < 0$. The logarithmic derivative tends to $+\infty$ at zero and to $1-n < 0$ at infinity. Thus there is a unique $t_n > 0$ satisfying

$$\coth t_n = n\tanh(nt_n). \tag{3}$$

The function ϕ_n increases on $(0, t_n)$, decreases on (t_n, ∞), and tends to zero at both ends. Define $b_{n,j} = 2\sin(j\pi/n)\phi_n(t_n)$, for $1 \leq j < n$.

Proposition 2. *For $n \geq 2$ and $\varepsilon > 0$, let $f_{n,\varepsilon}(z) = \varepsilon T_n(z) - 2\operatorname{Im} z$. Then*

$$N(f_{n,\varepsilon}) = n + 2\#\{1 \leq j < n : \varepsilon < b_{n,j}\} + \#\{1 \leq j < n : \varepsilon = b_{n,j}\}. \tag{4}$$

Every zero is nondegenerate unless $\varepsilon = b_{n,j}$ for some j. At such a parameter there is exactly one degenerate zero for each index j with $\varepsilon = b_{n,j}$.

Proof. On the real axis, $f_{n,\varepsilon}$ has the n simple zeros of T_n. At each such point x, its Jacobian is $\varepsilon^2 T_n'(x)^2 > 0$. On the upper branch $z_j(t)$ in (2), its real part vanishes precisely when $\varepsilon(-1)^j = 2\sin\theta_j\phi_n(t)$. On the reflected lower branch, the corresponding equation is $\varepsilon(-1)^j = -2\sin\theta_j\phi_n(t)$. Only the upper branch is possible for even j, and only the lower branch for odd j. On the selected branch, the equation reduces to

$$\varepsilon = 2\sin\theta_j\,\phi_n(t). \tag{5}$$

The monotonicity established before (3) gives two, one, or no solutions according as $\varepsilon < b_{n,j}$, $\varepsilon = b_{n,j}$, or $\varepsilon > b_{n,j}$. This proves (4).

The derivative of the imaginary part in the transverse θ-direction is, up to sign, $\varepsilon n(-1)^j\sinh(nt) \neq 0$. Along the selected branch, the derivative of the real part at a zero is a nonzero factor times $\phi_n'(t)$. In the

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regular coordinates (1), these derivatives prove nondegeneracy except at $t = t_n$. At t_n, the differential has rank one. Moreover, $\phi_n''(t_n) < 0$, because the logarithmic derivative is strictly decreasing. The two roots of (5) therefore meet in a quadratic tangency, and the stated degenerate zeros are the only ones. □

We now prove the explicit assertion of Theorem 1. The minimum of the $b_{n,j}$ is $b_{n,1}$, and

$$b_{n,1} \geq \frac{2\sin(\pi/n)\sinh(1/n)}{\cosh 1} \geq \frac{4}{n^2 \cosh 1} > \frac{1}{n^2}. \tag{6}$$

Here $\sin(\pi/n) \geq 2/n$, $\sinh(1/n) \geq 1/n$, and $\cosh 1 < 4$. Equations (4) and (6) show that $f_{n,1/n^2}$ has $n + 2(n-1)$ nondegenerate zeros. Finally, $-if_{n,1/n^2} = z - iT_n(z)/n^2 - \bar{z} = p_n(z) - \bar{z}$. Multiplication by $-i$ preserves zeros and nondegeneracy, completing the proof.

The location of the additional roots explains why the linearity of T_1 does not reduce the count. On every selected branch, one solution of (5) approaches $t = 0$, and the other approaches infinity as ε decreases to zero. Thus there are $n-1$ roots near the real critical points of T_n and $n-1$ roots escaping to infinity, in addition to its n real roots.

Figure 1 illustrates the count when $n = 5$.

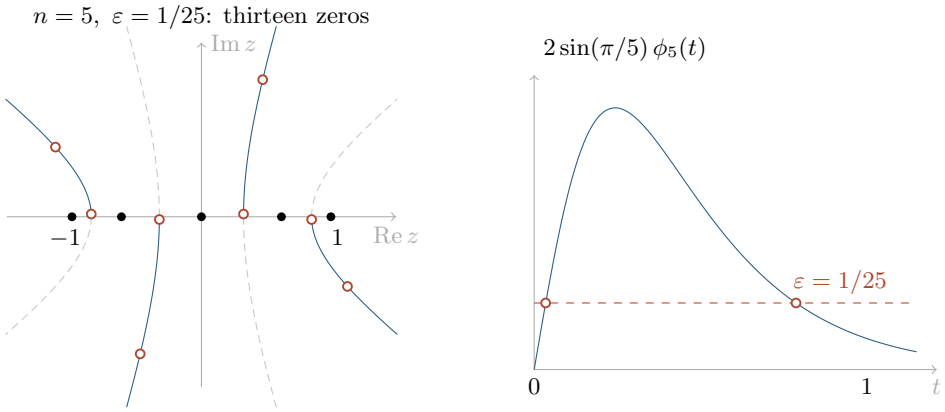


FIGURE 1. The degree-five example. Filled dots are the five real zeros; open circles are the eight nonreal zeros. Solid curves are the selected branches of $\operatorname{Im} T_5 = 0$; their conjugates are dashed. The right panel shows the two intersections that occur on each selected branch, here for $j = 1$ and $j = 4$.

REFERENCES

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